

Steam methane reforming in a microchannel reformer: Experiment, CFD-modelling and numerical study

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ABSTRACT

This paper is devoted to the investigation of the steam methane reforming process in a microchannel reformer over a Ni-based catalyst. The composition of reforming products and temperature at a reformer outlet was measured experimentally as well as with the help of CFD-modeling and numerical study. The presented new direct numerical method for modeling of steam methane reforming. The study of steam methane reforming was performed for the inlet temperatures (T_{in}) of 600–1000 K, for $Re = 10, 50, 100$, and heat flux through the walls of $q_w = 1000 \text{ W/m}^2$ and $q_w = 500 \text{ W/m}^2$. It was found that the methane conversion in the reformer decreases with increasing Reynolds number, heat flux through the wall, and inlet temperature. For $Re = 10$, $q_w = 1000 \text{ W/m}^2$ and $T_{in} = 1000 \text{ K}$ the methane conversion is 39% while equilibrium methane conversion is 63%. The temperature profiles in the reaction zone were determined. The temperature of the reaction mixture monotonically increases for all Reynolds numbers.

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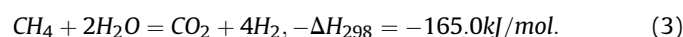
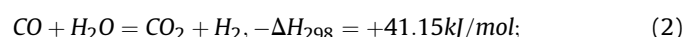
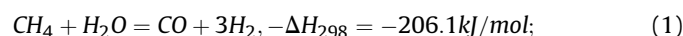
1. Introduction

Steam methane reforming is widely used in various industries: hydrogen production, chemical technology, energy, transport [1,2]. In addition, steam methane reforming has great prospects for use in cooling systems for heat-stressed structures. The method of cooling heat-stressed structural elements is widely discussed for use in the Ajax hypersonic aircraft [3]. In such cooling systems, methane reforming takes place in the microchannel reformers, the walls of which are activated with a thin layer of catalyst. The catalyst activates chemical reactions and suppresses external and internal diffusion resistance.

The steam methane reforming process in microchannels is well studied using experiments [4,5] and CFD modeling [6,7]. A large number of publications on steam methane reforming are due to a large number of practically interesting cases: different catalysts, different designs of reformers, different operating parameters, etc. Among the frequently encountered designs of reformers, two types can be distinguished: a fixed bed reaction system and a microchannel reaction system. Izquierdo et al. investigated the efficiency of the steam methane reforming process in the fixed-bed reaction systems and microchannel reaction systems that activated by Ni-

based (over MgO and Al_2O_3) and noble metal-based (Pd and Pt over Al_2O_3) catalysts [8]. The authors concluded that for all the catalysts, lower methane conversion and lower hydrogen yields were obtained in the fixed bed reactors at the same residence time. Also, higher methane conversions and higher hydrogen yields were obtained in the microchannel reactors.

The chemical reaction of steam methane reforming (SMR) is a well-studied process, both theoretically [9–14] and experimentally [15–19]. Various types of catalysts are used to increase the efficiency of this process. The nickel based catalysts are widely used in large-scale hydrogen industry [14,15,20,21]. In addition, various noble metals such as palladium (Pd) [22], platinum (Pt) [23], rhodium (Rh) [24], iridium (Ir) [9] and ruthenium (Ru) [25] have found a wide application in the catalysts for steam methane reforming as an active element. The steam methane reforming process can mainly be presented in three reversible reactions [15]: the endothermic reforming reactions (1) and (3), and the slightly exothermic water-gas shift reaction (WGS) (2):



The research object in this article is a microchannel reformer for

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steam methane reforming. Murphy et al. [26] presented the results of combined experimental and numerical investigation of the steam methane reforming process in a ceramic microchannel reformer. The authors determined that the heat-exchanger effectiveness of a microchannel reformer is up to 88% is experimentally demonstrated and 100% methane conversion is achieved at the reformer outlet. Murphy and co-workers concluded that microchannel technology has a positive outlook for use in industrial applications.

Cao et al. [27] reported the results of a numerical analysis on steam methane reforming in a plate microchannel reactor activated with a Ni-based catalyst. The authors investigated the effect of washcoat properties on the reforming and flow characteristics in the reaction space of the microchannel reformer. Based on the numerical analysis, the authors concluded that using a thicker washcoat carrying a greater loading of catalyst can considerably increase the reactor throughput within washcoat thickness of 75 μm , at the expense of certain loss of the catalyst productivity.

Jin et al. [28] experimentally investigated the microchannel methane steam reformers with improved heat transfer efficiency. The authors concluded that the heat transfer efficiency from a heat source to a catalyst is a much more important factor than the reaction activity of the catalyst itself on an industrial scale. In this investigation, the authors enhanced the heat transfer efficiency of the microchannel reactor for the steam methane reforming process by using porous-membrane-type catalysts.

The heat and mass transfer that occurs during chemical reactions in the microchannel reformers are characterized by low Reynolds numbers, large heat fluxes, and high rates of chemical reactions. Therefore, when calculating such reformers, it is necessary to take into account the longitudinal heat transfer and mixture components, especially in the initial section of the channel, where the maximum change in the properties of the reaction mixture occurs. In addition, in extended microchannels ($L/D > 10$), it is necessary to take into account the change in the transverse fluxes of heat and mixture components along the length of the microchannel due to a change in the reaction rate. Another feature of heat and mass transfer in the microchannel is a significant change in the physicochemical and thermophysical properties of the reaction mixture along its length.

Hsu reported the results of exact analysis of low Peclet number thermal entry region heat transfer in transversely nonuniform velocity fields [29]. The author determined that at low Peclet number the longitudinal thermal conduction has a significant effect on heat transfer at the microchannels with low L/D ratio. Hsu obtained a theoretical solution to the problem of thermal entry region heat transfer in the parallel-plate channel flow. However, Ksu considered in his model the constant thermophysical properties of the flow. Lui et al. [30]. Numerically investigated heat transfer in the microchannels when the thermophysical properties of the flow depend on temperature. The mentioned above papers show that the task of determining the parameters of heat and mass transfer is difficult.

An experimental study of heat and mass transfer in the microchannels of the reformer makes it possible to determine the main parameters of the process at the reformer outlet. However, it is rather difficult to determine the distribution of temperature, mass or molar fractions of components, and other parameters in the reaction space of a microchannel due to the size of the microchannel. In addition, experimental research, while providing accurate results, is costly. One of the ways to determine the characteristics of heat and mass transfer in the reaction channel of microreformers is CFD-modeling. The main advantage of CFD-modeling is obtaining visual results with minimal expenditure of money. At the same time, when verifying the results of CFD-

modeling, the problem of the reliability of the results obtained in the reformer microchannel arises, because as a rule verification occurs according to the flow parameters at the microreformer outlet. One of the ways to determine the parameters of heat and mass transfer in a microchannel is a conjugate numerical solution of the Navier-Stokes equations and the equations of chemical kinetics of the steam methane reforming. As a result of this solution, it becomes possible to determine temperature and mass flow for simple geometries of microchannels, which can be used, for example, to verify the results of CFD-modeling. The available scientific publications on the study of steam methane reforming in microchannels are based mainly on experimental studies and CFD-modeling or their combination. This article presents an experimental investigation, CFD-modeling via Ansys Fluent, and numerical study based on a conjugate solution of the Navier-Stokes equations and the equations of chemical kinetics of steam methane reforming in a microchannel reformer activated with a Ni-based catalyst.

2. Methods

2.1. Experiment

A schematic diagram for an experimental investigation of the steam methane reforming process is classical. Fig. 1 show the schematic diagram of an experimental setup. The microchannel reformers were produced by LLC "Catalheat". The reformer body was made of high alloy steel (30-30 Cr-Ni). The reformer has a rectangular body. The main dimensions are shown in Fig. 2. The central part (40 mm) of the reformer is activated by a catalyst. Heat is supplied only in the part of the reformer with a catalyst. The inlet and outlet areas of the reformer have adiabatic walls. An additional inlet area is used to form a developed flow at the inlet to the reaction space. The microchannel height is 600 μm . The thickness of the catalyst layer on each wall is 50 μm . The side walls are adiabatic. The heat source is an electric heater. Heat flow power is 1500 W/ m^2 .

This schematic diagram can be virtually divided into three blocks: first – preparation steam-methane mixture, second – methane reforming, third – measuring and storing of the results. To prepare the steam with a temperature of 400 °C and pressure of 1 bar, an electrical steam generator PLG-400 was used. Methane and nitrogen are stored in the cylinders. Nitrogen is used to preheat a catalyst bed of the reformer in a neutral atmosphere. An electrical heater of a steam-methane reforming mixture is used to preheat a reaction mixture up to temperature in the reformer. High purity methane N99 is used as a reagent for reforming in the experiments, as well as high purity water is used for steam generation. The temperature of each stream in this schematic diagram is controlled by the tungsten-rhenium alloy thermocouples. The synthesis gas composition was determined with a help of the portable infrared gas analyzer Gasboard 3100 P, which can be connected to a personal computer. The photos of the gas analyzer and a catalyst plate are presented in Fig. 1.

Before the experiment, the electrical furnace heats the reaction space of the reformer to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. The heating of the catalyst takes approximately 2–3 min. The fresh microreformers are used for every 5 experiments.

The plate reformer is activated by a commercial Ni-based catalyst. For experiment the Ni–Al₂O₃ catalyst is used with the following chemical composition: NiO = 14.5%; SiO₂ = 0.2%; support CaO–MgO–La₂O₃– α -Al₂O₃. Other properties are presented below: bulk density is 680 kg/m³; surface area is 4.5 m²/g_{cat}; average

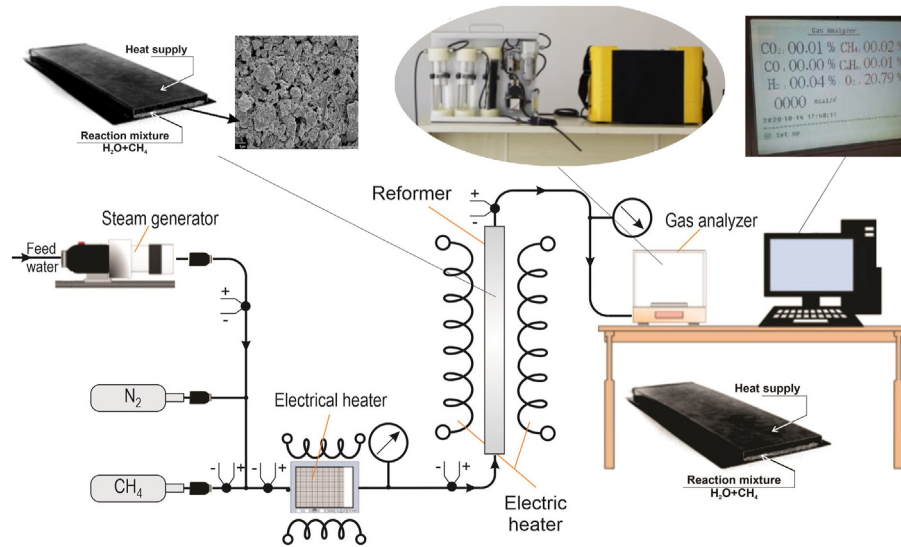


Fig. 1. Schematic diagram of an experimental setup.

porosity of a catalyst is about 41%. The morphology of a catalyst is presented in Fig. 1.

2.2. CFD-modeling

2.2.1. Geometry and mesh

A laminar flow of a multicomponent mixture of methane, steam and products of their reactions in a two-dimensional flat micro-channel is considered, taking into account chemical reactions on the walls of the microchannel. When chemical reactions take place, the temperature and molar fractions of the components of the gas mixture change. There is no catalyst at the inlet of the micro-channel. The chemical reactions take place in presence of the catalysts when external heat is supplied to the walls of the microchannel. The supplied heat is absorbed during endothermic reactions, the rest of the heat is spent on heating the mixture.

The schematic view of a computational domain for CFD-modeling is presented in Fig. 2. The mesh structure is shown in Fig. 3. The geometric characteristics are the same as in the experiments. To simplify the CFD-model, in this paper 2D model was considered.

2.2.2. Governing equations

In this CFD-model the mass, momentum and energy conservation equations in combination with chemical model of the SMR process are solved in ANSYS Fluent. The following assumptions are used for this CFD-model:

- steam methane reforming is steady-state process;
- no the flux of energy due to a mass concentration gradient (no Dufour effects) [31];
- work by viscous forces and by pressure is not done;
- any other surface oxidation reactions (for example, metal oxidation reactions) are absent.

The differential equation for mass conservation for steady-state conditions for 2D-geometry is following:

$$\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) = 0 \quad (4)$$

In these equations v_x and v_y are the axial and radial velocity,

respectively; x and y are the axial and radial coordinate, respectively.

The moment conservation equation in an inertial reference frame can be written in the following form [32]:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot \vec{p} + \nabla \cdot (\vec{\tau}) + \rho \vec{g} + \vec{F} \quad (5)$$

where p is the static pressure; $\rho \vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. The stress tensor $\vec{\tau}$ is given by follows equation:

$$\vec{\tau} = \mu \left[\left(\nabla \vec{v} + \vec{v}^T \right) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (6)$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The equation for energy conservation for the reaction zone has following form:

$$\nabla \cdot \vec{u} (p + \rho \cdot h_f) = \nabla \cdot [k_{eff} \cdot \nabla T - \left(\sum h_j \cdot \vec{J}_j \right) + \vec{\tau} \cdot \vec{v}] + S_f^h \quad (7)$$

In this equation T is the temperature of reaction mixture in the reaction space; ρ is the density; h_f is enthalpy of reaction mixture; $\vec{J}_j$ is the diffusion flux of j component; k_{eff} is the conductivity; h_j is the enthalpy of j component; S_f^h is the fluid enthalpy source term (include the source of energy due to chemical reaction).

The equation of energy conservation for the reformer wall with a catalyst can be described by follow equations:

$$\nabla \cdot (k_w \cdot \nabla T) = q_w \quad (8)$$

where k_w is the thermal conductivity of the metal wall; q_w is heat through the metal wall.

The energy conservation equation for the wall without a catalyst can be written as:

$$\nabla \cdot (k_w \cdot \nabla T) = 0. \quad (9)$$

The transport processes of j component of gas mixture can be described by following equation:

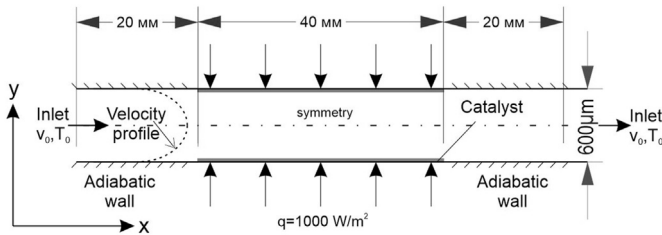


Fig. 2. Schematic view of a microchannel reformer.

$$\nabla \cdot (\rho \cdot \vec{v} \cdot Y_j) = -\nabla \cdot \vec{J}_j + R_j \quad (10)$$

where R_j is the reaction rate of formation or consumption of j th component; Y_j is the mass fraction of j th component of gas mixture; $\vec{J}_j$ is diffusion flux of j component.

The diffusion processes of j th component can be expressed as:

$$\vec{J}_j = -\left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t} \nabla Y_j - D_{Tj} \frac{\nabla T}{T}\right) \quad (11)$$

where $D_{j,m}$ and D_{Tj} are the mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

Usually, SMR occurs in the laminar flow of the reaction gas mixture. Therefore *laminar* is used in Fluent solver.

The reaction rate of the methane consumption through reactions (1 and 3) can be found from following expression:

$$r_{CH_4} = R_1 + R_3. \quad (12)$$

The reaction rate of carbon monoxide (CO) and carbon dioxide (CO₂) production can be found from following expressions:

$$r_{CO} = R_1 - R_2. \quad (13)$$

$$r_{CO_2} = R_3 + R_2. \quad (14)$$

The reaction rate for each reaction can be calculated by following equations [33]:

$$R_1 = \frac{k_1}{p_{H_2}^{2.5}} \left(p_{CH_4} \cdot p_{H_2O} - \frac{p_{H_2}^3 \cdot p_{CO}}{K_{e1}} \right) \cdot \frac{1}{Q_r^2}, \quad (15)$$

$$R_2 = \frac{k_2}{p_{H_2}} \left(p_{CO} \cdot p_{H_2O} - \frac{p_{H_2} \cdot p_{CO_2}}{K_{e2}} \right) \cdot \frac{1}{Q_r^2}, \quad (16)$$

$$R_3 = \frac{k_3}{p_{H_2}^{3.5}} \left(p_{CH_4} \cdot p_{H_2O}^2 - \frac{p_{H_2}^4 \cdot p_{CO_2}}{K_{e3}} \right) \cdot \frac{1}{Q_r^2}, \quad (17)$$

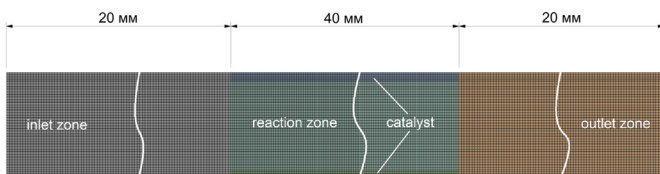


Fig. 3. The computational domain and mesh structure.

$$Q_r = 1 + K_{CO} \cdot p_{CO} + K_{H_2} \cdot p_{H_2} + K_{CH_4} \cdot p_{CH_4} + \frac{K_{H_2O} \cdot p_{H_2O}}{p_{H_2}}, \quad (18)$$

where p_i - partial pressure of i reacting element, bar; K_{je} - equilibrium constant of 1, 2, 3 reactions, respectively; k_j - kinetic rate constant of j reaction.

In Eq. (18) adsorption constants of i element (K_i) can be determined as:

$$K_i = K_{0i} \cdot e^{-\Delta H_i / (R_0 T)}, \quad (19)$$

where H_i - adsorption enthalpy of i element, J/mol; K_{0i} - constant of i component.

2.2.3. Boundary conditions

The specific heat and density of the gas mixture in the reaction zone are calculated by the equations of mixing-law and incompressible-ideal-gas law, respectively. It can be possible, because the temperature of reaction mixture is moderate high (less than 1000 K) and the pressure in the reaction space is moderate low (less than 10 bar). The matching models are selected in *Fluent* solver. The pressure-based solver is chosen. Therefore, the governing equations are solved sequentially. The convergence criterion for momentum, continuity and species are defined as 10^{-3} , for energy less than 10^{-6} . The hybrid initialization is used to initialize calculation. For described equations the Second-Order Upwind approximation method is used to obtain a higher calculation accuracy. When second-order accuracy is desired, quantities at cell faces are computed using a multidimensional linear reconstruction approach.

Moreover, the boundary conditions for the computational domain of the reformer shown in Fig. 2 are determined as follows:

- reformer inlet:

$$x = 0; T_g = T_{g(in)}; C_i = C_{i(in)}. \quad (20)$$

- reformer outlet:

$$x = L; \frac{\partial C_i}{\partial x} = 0. \quad (21)$$

- reformer center:

$$y = 0; \frac{\partial C_i}{\partial y} = 0; \frac{\partial T_g}{\partial y} = 0. \quad (22)$$

where C_i is mole concentration of i element.

2.3. Numerical study

For laminar flow, the Navier-Stokes equations can be solved numerically with high accuracy because there is no turbulence. To eliminate pressure from the Navier-Stokes equations, these equations were written in vorticity and stream function variables:

$$\frac{\partial}{\partial x'} \left(\frac{1}{\rho'} \frac{\partial \psi'}{\partial x'} \right) + \frac{\partial}{\partial y'} \left(\frac{1}{\rho'} \frac{\partial \psi'}{\partial y'} \right) + \omega' = 0 \quad (23)$$

In Eq. (24) the following notations are used:

$$\begin{aligned} & \frac{Re}{2} \left[\frac{\partial}{\partial x'} \left(\frac{\partial \psi'}{\partial y'} \omega' \right) - \frac{\partial}{\partial y'} \left(\frac{\partial \psi'}{\partial x'} \omega' \right) - \frac{1}{2} \left(\frac{\partial \rho'}{\partial y'} \frac{\partial (U'^2 + V'^2)}{\partial x'} - \frac{\partial \rho'}{\partial x'} \frac{\partial (U'^2 + V'^2)}{\partial y'} \right) \right] = \\ & = \frac{\partial^2}{\partial x'^2} (\mu' \omega') + \frac{\partial^2}{\partial y'^2} (\mu' \omega') - 2 \left(\frac{\partial}{\partial x'} \left(\frac{\partial \mu'}{\partial y'} \frac{\partial U'}{\partial x'} - \frac{\partial \mu'}{\partial x'} \frac{\partial U'}{\partial y'} \right) + \frac{\partial}{\partial y'} \left(\frac{\partial \mu'}{\partial y'} \frac{\partial V'}{\partial x'} - \frac{\partial \mu'}{\partial x'} \frac{\partial V'}{\partial y'} \right) \right) \end{aligned} \quad (24)$$

$$U' = \frac{1}{\rho'} \frac{\partial \psi'}{\partial y'}, V' = -\frac{1}{\rho'} \frac{\partial \psi'}{\partial x'}, \omega' = \frac{\partial V'}{\partial x'} - \frac{\partial U'}{\partial y'} \quad (25)$$

The energy conservation equation for the reaction zone can be written as following:

The species transport equation for i -th element can be written as following:

$$G_{ik} = \frac{1}{2\sqrt{2}} \frac{(1 + (\mu_i/\mu_k)^{1/2} (M_k/M_i)^{1/4})^2}{(1 + M_i/M_k)^{1/2}}, \quad (30)$$

$$A_{ik} = \frac{1}{2\sqrt{2}} \frac{(1 + (\lambda_i/\lambda_k)^{1/2} (M_i/M_k)^{1/4})^2}{(1 + M_i/M_k)^{1/2}} \quad (31)$$

The diffusion and mixture enthalpy was calculated as following:

$$\begin{aligned} & \frac{Re}{2} \left(\frac{\partial (\rho' U' h')}{\partial x'} + \frac{\partial (\rho' V' h')}{\partial y'} \right) = \frac{1}{Pr} \left[\frac{\partial}{\partial x'} \left(\lambda' \frac{\partial T'}{\partial x'} \right) + \frac{\partial}{\partial y'} \left(\lambda' \frac{\partial T'}{\partial y'} \right) \right] + \\ & + \frac{Re}{2} \frac{R_0}{C_{P_0}} \left(U' \frac{\partial P'}{\partial x'} + V' \frac{\partial P'}{\partial y'} \right) + M^2 \frac{R_0}{C_{P_0}} \mu' \left[\left(\frac{\partial U'}{\partial y'} + \frac{\partial V'}{\partial x'} \right)^2 + 4 \left(\frac{\partial U'}{\partial x'} \right)^2 + 4 \left(\frac{\partial V'}{\partial y'} \right)^2 \right] + \\ & + \frac{1}{Sc} \left[\frac{\partial}{\partial x'} \left(\sum_i h'_i \rho' D' \frac{\partial Y'_i}{\partial x'} \right) + \frac{\partial}{\partial y'} \left(\sum_i h'_i \rho' D' \frac{\partial Y'_i}{\partial y'} \right) \right] \end{aligned} \quad (26)$$

$$\frac{Re}{2} Sc \left(\frac{\partial (\rho' U' Y'_i)}{\partial x'} + \frac{\partial (\rho' V' Y'_i)}{\partial y'} \right) = \frac{\partial}{\partial x'} \left(\rho' D' \frac{\partial Y'_i}{\partial x'} \right) + \frac{\partial}{\partial y'} \left(\rho' D' \frac{\partial Y'_i}{\partial y'} \right) \quad (27)$$

In Eqs. (24)–(27) the following notations are used: U, V - longitudinal and transverse velocity components; ψ - Stokes flow function; h - enthalpy of gas mixture; Re - Reynolds number; Pr - Prandtl number; Sc - Schmidt number; M - Mach number; R_m - molar gas constant; Y_i - mass fraction of i -th element; M_i - molar mass.

The Mach number in the reaction space of the reformer is lower than 0.001 and pressure drop in the reformer are negligible. Therefore the gas equation has the following form:

$$1 = \frac{\rho' R_m T'}{R_0} \sum \frac{Y_i}{M_i} \quad (28)$$

The viscosity (μ) and heat conductivity (λ) depends on temperature and molar fraction of i -th element. Therefore viscosity (μ) and heat conductivity (λ) are determined as following:

$$\mu = \sum_{i=1}^n \frac{\mu_i X_i}{\sum_{k=1}^n G_{ik} X_k}, \lambda = \sum_{i=1}^n \frac{\lambda_i X_i}{\sum_{k=1}^n A_{ik} X_k} \quad (29)$$

In Eq. (29) G_{ik} and A_{ik} factors are determined as:

$$D = \frac{9,99 \cdot 10^{-5} T^{1,75}}{P}, \quad (32)$$

$$h = \sum_i^N Y_i h_i, h_i = h_{i,0} + \int_{T_s}^T C_{P_i} dT. \quad (33)$$

where $h_{i,0}$ is formation enthalpy of i th component at $T = 298$ K.

The boundary condition for the wall have the following form:

$$-\rho' D'_i \frac{\partial Y'_i}{\partial y'} = Da_i s'_i, -\lambda' \frac{\partial T'}{\partial y'} = \sum_i \frac{s'_i h'_i Da_i Pr}{Sc} + q'_w \quad (34)$$

where s_i - total molar flux of i -th element from the wall to reaction space; Da_i - Damkohler number; q_w - heat flux through the wall.

In this numerical calculation the chemical kinetics were the same as in the CFD-model realized via Ansys Fluent in section (2.2).

3. Results and discussion

The advantage of this study is the results obtained for one type of reformer in three different ways: experiment, CFD-modelling and numerical calculation.

To carry out SMR process, an external heat to the reaction zone must be supplied because the reactions of SMR are strongly endothermic. In this study, the external heat is supplied evenly to the reaction space through the wall. The heat flux density is

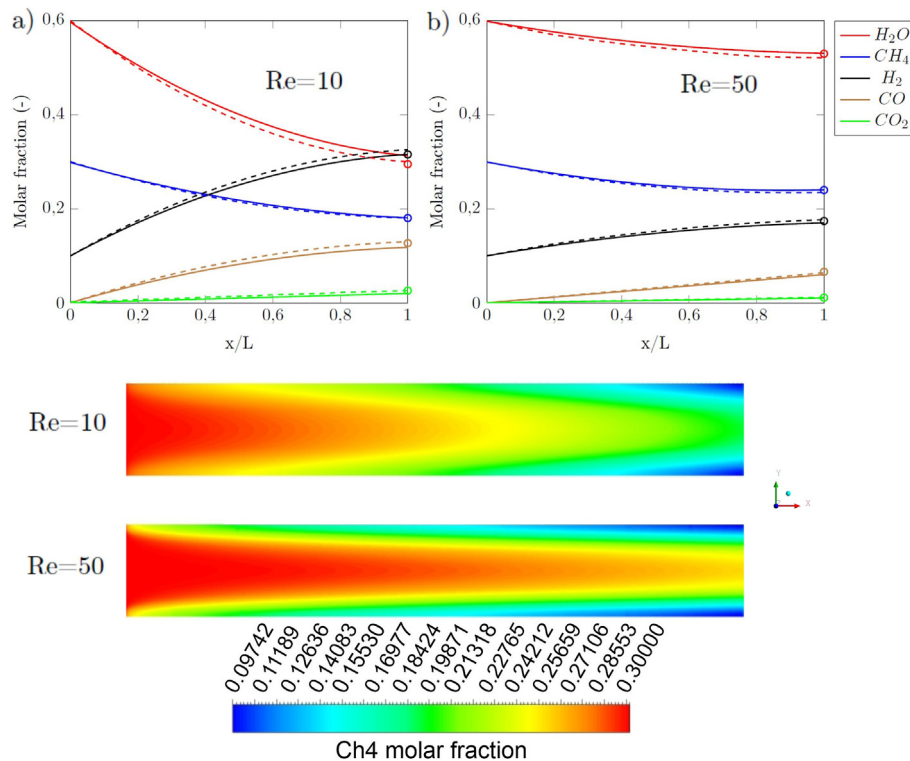


Fig. 4. Average molar fraction of syngas species along the length of the reformer for $q_w = 1000 \text{ W/m}^2$ and $T_{in} = 1000 \text{ K}$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results. (These contours were scaled with the following parameters: $x - 0.3$; $y - 3.0$).

1000 W/m^2 and 500 W/m^2 . Part of this heat flux is spent on the endothermic reaction, the rest part is spent on heating the reaction mixture.

The investigation is performed for inlet temperature of 800 K , 900 K and 1000 K at $Re = 10$, $Re = 50$, and $Re = 100$. The inlet reaction has the following molar composition: $\text{CH}_4:\text{H}_2\text{O}:\text{H}_2 = 0.3:0.6:0.1$.

3.1. Syngas composition

The syngas compositions in the reformer outlet were experimentally determined. Also, the syngas compositions were calculated via CFD-model and numerical model for all reaction space. Fig. 4 shows the distribution of the molar fraction of syngas species along the length of the reformer determined with the help of CFD-modeling (solid lines and CH_4 molar fraction contours), the numerical calculation (dashed lines) and experimental investigation (hollow points) for $Re = 10$ and $Re = 50$ at $q_w = 1000 \text{ W/m}^2$ and $T_{in} = 1000 \text{ K}$. Moreover, Fig. 4 presents the contours of methane (CH_4) mass fraction for $Re = 10$ and $Re = 50$. For clarity, the contours are scaled along the axis Y by five times.

The average molar fractions along the length in Fig. 4 were obtained for each section of the reformer. The experimental results were obtained after mixing the syngas after the reformer outlet. Fig. 4 shows that the molar fraction of the syngas species along the reformer length is changed monotonically. The molar fraction of H_2O and CH_4 is decreasing while the molar fraction of H_2 , CO and CO_2 is increasing. Moreover, Fig. 4 make it is clear that increasing in

Reynolds number leads to a decrease in the methane conversion and syngas yield.

Fig. 5 show the distribution of the molar fraction of syngas species along the length of the reformer determined with the help of CFD-modeling (solid lines and CH_4 molar fraction contours), the numerical calculation (dashed lines) and experimental investigation (hollow points) for $Re = 10$ and $Re = 50$ at $q_w = 500 \text{ W/m}^2$ and $T_{in} = 1000 \text{ K}$. Fig. 5 make it is clear that decreasing in the heat flux through the wall leads to decreasing in the methane conversion and hydrogen and carbon monoxide yield. This is due to the fact that the steam methane reforming process is strongly endothermic and to get the high methane conversion the required heat must be constantly supplied to reaction space.

To compare the reforming characteristics of various cases the methane conversion was determined. Moreover, based on the author's previous paper [34], the equilibrium methane conversion was calculated. The methane conversion is determined as follows:

$$X_{\text{CH}_4}(\%) = \frac{[\text{CH}_4]_{in} - [\text{CH}_4]_{out}}{[\text{CH}_4]_{in}} \times 100\% \quad (35)$$

Fig. 6 shows the methane conversion obtained for various cases - various inlet temperature and various heat flux through the wall. Also, the equilibrium methane conversions were added as a limit value in the methane conversion. The methane conversion for Fig. 6 obtained in this study is determined as an averaging value of results calculated with the help of CFD-model, numerical calculation, and experiment. Fig. 6 depicts that an increase in heat flux increases methane conversion. On the other hand, an increase in the

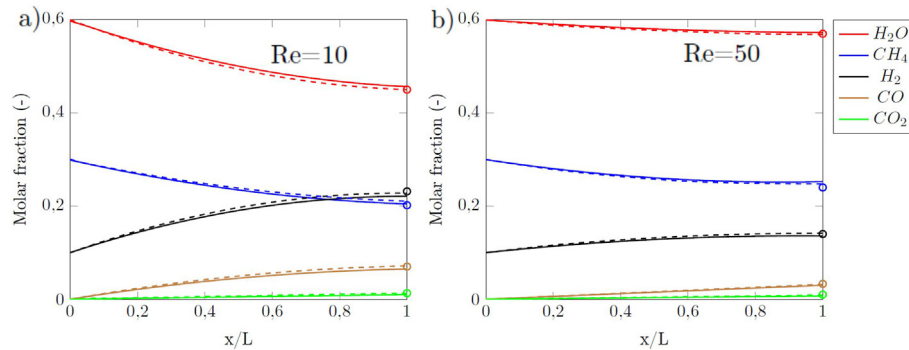


Fig. 5. Average molar fraction of syngas species along the length of the reformer for $q_w = 500 \text{ W/m}^2$ and $T_{in} = 1000\text{K}$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results.

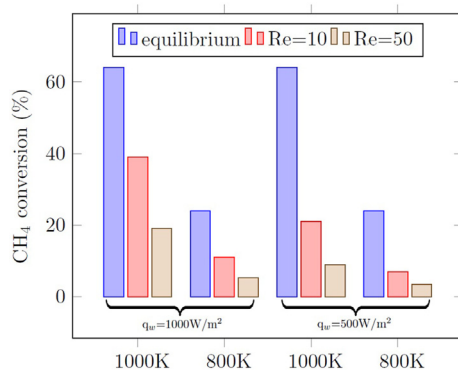


Fig. 6. The comparison of the methane conversion.

Reynolds number leads to a decrease in methane conversion. This is because the residence time decreases. It is known that a decrease in the contact time leads to a decrease in the conversion of methane [15,33].

Carbon formation is another factor that needs to be considered especially at high temperatures. However, there is sufficient information in literature that carbon formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as a steam-to-methane ratio [35]. In this study, steam-to-methane ratio in all different feed gas compositions is either equal to or greater than unity, which indicates that there is enough steam to oxidize and prevent formation of solid carbon. This observation is in a good agreement with the authors' previous study on carbon formation during steam methane reforming [36].

3.2. Temperature

Temperature is an important parameter to be controlled in micro reformers. Fig. 7 show the temperature distribution in the reaction zone for $T_{in} = 1000\text{K}$ and $q_w = 1000 \text{ W/m}^2$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results.

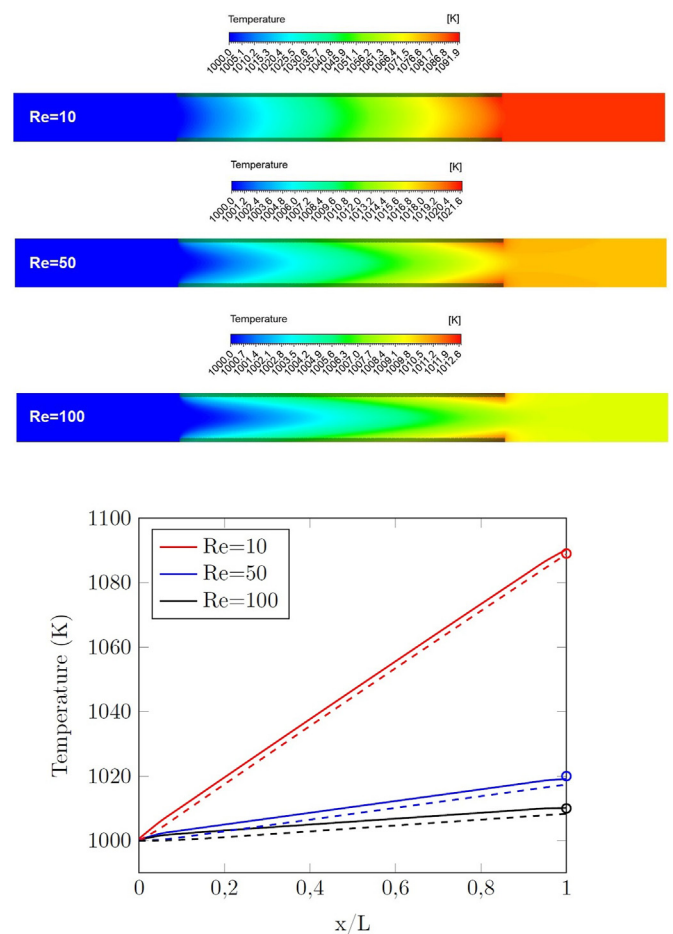


Fig. 7. The temperature profiles in the reaction zone for $T_{in} = 1000\text{K}$ and $q_w = 1000 \text{ W/m}^2$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results. (These contours were scaled with the following parameters: $x - 0.3$; $y - 3.0$).

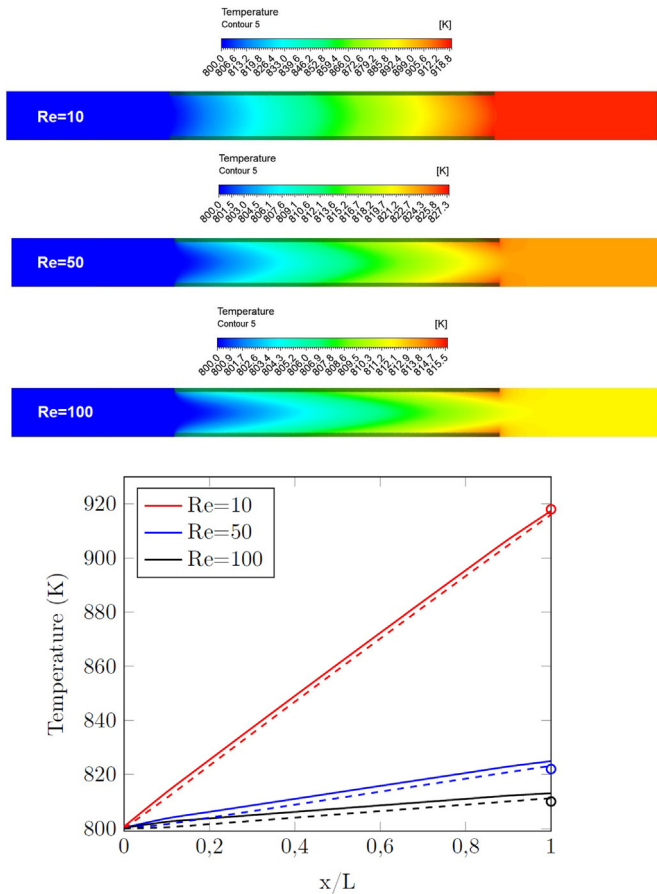


Fig. 8. The temperature profiles in the reaction zone for $T_{in} = 800\text{K}$ and $q_w = 1000\text{ W/m}^2$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results. (These contours were scaled with the following parameters: $x - 0.3$; $y - 3.0$).

Fig. 7 depicts that temperature in the reaction zone monotonically increases for all Reynolds numbers. This is due to the fact that the reaction rate of steam methane reforming is almost constant along the entire length of the reaction zone. As a result, the endothermic effect of the steam methane reforming reaction is constant. Therefore, part of the heat from the wall used to heat the reaction mixture is also constant. We calculated that 73% of the heat from the wall is consumed by the steam methane reforming reaction and 27% of the heat from the wall is used for the reaction mixture heating. However, a decrease in temperature at the reaction zone inlet leads to a decrease in the reaction rate of the steam methane reforming reaction and, as a consequence, to a decrease in the proportion of heat from the wall, which is spent on the methane reforming reaction. Therefore, with decreasing inlet temperature, the temperature of the reaction mixture will increase more noticeably.

Fig. 8 shows the temperature profiles in the reaction zone for $T_{in} = 800\text{K}$ and $q_w = 1000\text{ W/m}^2$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results. The temperature of the reaction mixture monotonically increases for all Reynolds numbers. However, the

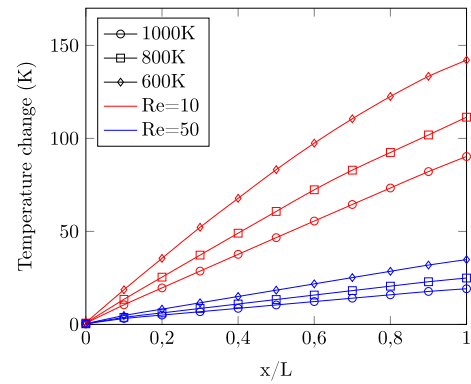


Fig. 9. The temperature profiles in the reaction zone for $T_{in} = 800\text{K}$ and $q_w = 1000\text{ W/m}^2$: solid lines – CFD-modeling; dashed lines – numerical calculation; hollow points – experimental results.

temperature increasing for $Re = 10$ up to 100 K leads to an increase in the reaction rate. Therefore, the temperature profile for $Re = 10$ is not a straight line. In the temperature range above 850 K , there is a subtle bending of the profile. The temperature profiles for $Re = 50$ and $Re = 100$ is a straight line because the reaction rate, and as a consequence, the endothermic effect of the steam methane reforming reaction is almost constant for the reaction zone. We calculated that 28% of the heat from the wall is consumed by the steam methane reforming reaction and 72% of the heat from the wall is used for the reaction mixture heating for $Re = 50$ and $Re = 100$. And 34% of the heat from the wall is consumed by the steam methane reforming reaction and 66% the heat from the wall is used for the reaction mixture heating for $Re = 10$.

As the temperature in the reaction zone increases, the reaction rate increases. If the temperature changes insignificantly, then the reaction rate changes minimally. However, for $Re = 10$, the temperature increases significantly especially for the inlet temperatures of 800 K and 600 K . This is due to the fact that the reaction rate is minimal and, as a consequence, the endothermic effect of the steam methane reforming reaction is also minimal. Therefore, all the heat from the walls is mainly spent on heating the reaction mixture. However, as the temperature in the reaction zone increases, the reaction rate increases. In this case, an increase in the reaction rate decreases the rate of temperature increase. Fig. 9 shows the temperature change for 800 K and 600 K . Fig. 9 shows that for the inlet temperature of 600 K and $Re = 10$, the rate of the temperature change decreases when the temperature of the reaction mixture increases.

To assess the accuracy of the experimental results of the outlet temperature, an analysis of the error distribution is performed. A histogram of the error distribution of the experimental results is shown in Fig. 10. Fig. 10 shows that the most accurate expressions are obtained for $Re = 50$.

The statistical analysis by the Pearson criterion, performed on the basis of the 9-bit histogram in Fig. 10, showed that the actual error distribution of the experimental results for $Re = 50$ and $q_w = 1000\text{ W/m}^2$ is in good agreement with the law of the normal Gaussian distribution ($\chi_{func} = 4.09 < \chi_{exp} = 12.592$, with a confidence level of $P = 95\%$ and the number of degrees of freedom

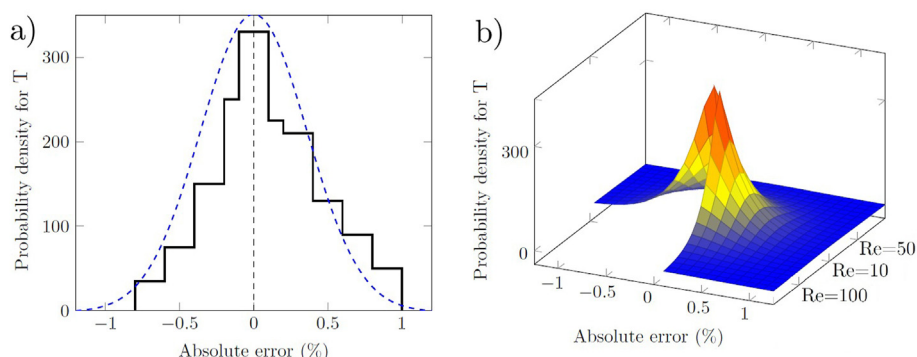


Fig. 10. Histogram of error distribution of the approximating functions: a) $T = 700\text{ }^{\circ}\text{C}$, $\text{H}_2\text{O} = 0.0$ (dashed line – Gaussian curve; solid line – histogram); b) Fixed $T = 700\text{ }^{\circ}\text{C}$ for $\text{H}_2\text{O} = 0.0 \dots 2.0$.

$m = 6$). Therefore, the experimental results are a high accuracy.

4. Conclusions

This work presents the results of the experimental and numerical investigations as well as CFD-modeling of the steam methane reforming process in a microchannel reformer with a reaction zone activated by a Ni-based catalyst ($\text{Ni}/\alpha\text{-Al}_2\text{O}_3$). A kinetic model for the CFD-model using a global mechanism of LHHW kinetics was taken from the literature. It was found that the methane conversion in the reformer decreases with increasing Reynolds number, heat flux through the wall, and inlet temperature. For $\text{Re} = 10$ and T_{in} the methane conversion is 39.2% ($q_w = 1000\text{ W/m}^2$) and 21.4% ($q_w = 500\text{ W/m}^2$) while equilibrium methane conversion is 63.4%. For $\text{Re} = 50$ and T_{in} the CH_4 conversion is 19.1% ($q_w = 1000\text{ W/m}^2$) and 4.7% ($q_w = 500\text{ W/m}^2$). The molar fraction of the syngas species along the reformer length is changed monotonically. Also, the temperature profiles in the reaction zone were determined. The temperature of the reaction mixture monotonically increases for all Reynolds numbers at $T_{in} = 800\text{ K}$ and $T_{in} = 1000\text{ K}$. For $T_{in} = 600\text{ K}$ and $\text{Re} = 10$, the rate of the temperature change decreases when the temperature of the reaction mixture increases. We calculated that 73% of the heat from the wall is consumed by the steam methane reforming reaction and 27% of the heat from the wall is used for the reaction mixture heating for $T_{in} = 1000\text{ K}$ and $q_w = 1000\text{ W/m}^2$ for $\text{Re} = 10, 50, 100$. For the inlet temperature of 600 K and $\text{Re} = 10$ for $q_w = 500\text{ W/m}^2$, the rate of the temperature change decreases when the temperature of the reaction mixture increases.

Credit author statement

Methodology, Writing, Experiment Pashchenko Dmitry; CFD-modeling - Mustafin Ravil; Visualization, Writing- Mustafina Anna. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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